one unique rate of gas infall, not too high and not too low, at which the X-rays and gas are in mutual equilibrium.

This picture of the flow of gas and X-rays disturbed me. I knew full well that if, on Earth, one tries to hold a dense fluid such as liquid mercury up by means of a less dense fluid such as water below it, tongues of mercury quickly eat their way down into the water, the mercury goes whooshing down, and the water goes whooshing up (Figure 6.4b). This phenomenon is called the *Rayleigh–Taylor instability.* In Zel'dovich's picture, the X-rays were like the low-density water and the infalling gas was like the high-density mercury. Wouldn't tongues of gas eat their way into the X-rays, and wouldn't the gas then fall freely down those tongues, destroying Zel'dovich's self-regulated flow (Figure 6.4c)? A detailed calculation with the laws of physics could tell me whether this happens, but such a calculation would be very complex and time consuming; so, rather than calculate, I asked Zel'dovich one afternoon in 1969, when we were discussing physics in his apartment in Moscow.

Zel'dovich looked a bit uncomfortable when I raised the question, hut his answer was firm: "No, Kip, that doesn't happen. There are no tongues into the X-rays. The gas flow is stable." "How do you know, Yakov Borisovich?" I asked. Amazingly, I could not get an answer. It seemed clear that Zel'dovich or somebody had done a detailed calculation or experiment showing that X-rays can push hard on gas without Rayleigh–Taylor tongues destroying the push, but Zel'dovich could not point me to any such calculation or experiment in the published literature, nor would he describe for me the detailed physics that goes on. How uncharacteristic of him!

A few months later I was hiking in the high Sierras in California with Stirling Colgate. (Colgate is one of the best American experts on the flows of fluids and radiation, he was deeply involved in the late stages of the American superbomb effort, and he was one of the three Livermore physicists who had simulated a star's implosion on a computer.) As we hiked, I posed to Colgate the same question I had asked of Zel'dovich, and he gave me the same answer: The flow is stable; the gas cannot escape the force of the X-rays by developing tongues. "How do you know, Stirling?" I asked. "It has been shown," he replied. "Where can I find the calculations or experiments?" I asked. "1 don't know ..."